

Aditya Kulkarni

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Raleigh, NC (remote or open to SF / NYC)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience designing and shipping agentic AI systems and RAG pipelines in Python and TypeScript. At Isomer AI, built multi-model LLM agent tooling across Claude, GPT-4o, and Gemini using LangChain-compatible MCP orchestration, running evaluations across 8+ model variants in production. Independently built AlgoPulse (agentic orchestration at 2,000 async inference req/sec with Temporal.io and Kafka) and Glance AI (per-tenant RAG with deterministic outputs for fintech/healthcare). Startup-native, comfortable owning delivery end-to-end from prototype to production. Case studies: kulkarniaditya.com

CORE COMPETENCIES

Agentic Workflow Design

RAG Pipeline Design & Optimization

LangChain / LangGraph

Multi-model LLM Routing

Prompt Engineering & Evaluation

Python & TypeScript

Vector Databases

Enterprise AI Delivery

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Designed and shipped production agentic workflows using **LLM APIs across 3 providers** (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models with MCP orchestration -- multi-step agent pipelines with tool use, prompt engineering, and real-time evaluation in a live claims automation product
- Built Temporal.io workflow orchestration for durable agent execution, real-time event pipelines (Pusher), and document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3) -- **394 commits, 7 production deployments**
- Ran multi-model evaluations to tune agent output quality and reliability; integrated Sentry observability to monitor agent behavior in production across TypeScript / Node.js / PostgreSQL / Redis stack

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed cloud-based genomic data platform (500GB, 100+ researchers worldwide) on Google Cloud Run; applied hyperparameter tuning and time-series ML analysis improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built AI-driven automation workflows for enterprise insurance clients using OCR, NLP, and ML pipelines; improved claim accuracy by 25% across production deployments on AWS and GCP
- Supported model validation, red-teaming, and reliability improvements with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed data pipelines (Hadoop, Spark, Kafka) for a support ticket analysis system; reduced processing latency by 20%

PROJECTS

Glance AI -- Privacy-First RAG Pipeline Independent, 2025

Designed and built a production-ready RAG system for fintech and healthcare environments with per-tenant data isolation at the vector retrieval layer. Prevents cross-tenant context leakage; deterministic output architecture eliminates LLM hallucinations in regulated environments. Covers the full RAG stack: chunking strategy, vector DB scoping, retrieval optimization, and output evaluation.

Python, LLMs, Vector Databases, RAG, Prompt Engineering

AlgoPulse -- Agentic AI Orchestration Platform Independent, 2025

Identified API-polling bottleneck in a multi-agent orchestration system under load; re-architected to event-driven streaming with Supabase Edge Functions, Temporal.io durable workflows, and Kafka. Scaled to **2,000 async inference requests/second** with zero message loss. Built reusable agent patterns for fault tolerance and state recovery.

Python, TypeScript, Temporal.io, Kafka, Supabase Edge Functions

PerCent -- GenAI Finance App HackNC24 Top 10 / 100+ teams

Built a RAG-based conversational AI system with Qwen-14B-Chat for financial domain reasoning -- goal tracking, expense logging, multi-step chatbot. Shipped complete MVP in 48 hours; ranked Top 10 of 100+ teams. Demonstrates rapid agentic prototyping to production-ready delivery.

Python, Hugging Face, Qwen-14B-Chat, RAG, LangChain

Mapify -- ML and NLP Pipeline IEEE Published

T5-transformer pipeline converting 1,500+ PDFs into structured mind maps and flowcharts; reduced review time by 60%. IEEE published, presented to 500+ attendees. Demonstrates NLP pipeline design and evaluation methodology from scratch.

T5-transformer, NLTK, NetworkX, Python

EDUCATION

Master of Science, Computer Science North Carolina State University

Aug 2024 - May 2026

Courses: Artificial Intelligence, Machine Learning, Data Science, Algorithms, Software Engineering

B.Tech, Computer Science Mumbai University

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Machine Learning, Big Data Infrastructure, Blockchain, Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C

AI/ML: LLM APIs (Claude, GPT-4o, Gemini), LangChain, LangGraph, MCP, RAG, Vector Databases, Hugging Face, TensorFlow, PyTorch, Prompt Engineering

Agent Infra: Temporal.io, Kafka, Supabase Edge Functions, Distributed Systems, Async Programming

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry **Frameworks:** Node.js, React, Express, Prisma, PostgreSQL, Redis